

S 1143339 FU-LISYUN

(1)

a.

a	b	Not_a	Not_b	NAND (output)
0	0	1	1	0
1	0	0	1	1
0	1	1	0	1
1	1	0	0	1

$\Rightarrow$ OR

b.

a	b	AND	NOR	NOR (output)
0	0	0	1	0
1	0	0	0	1
0	1	0	0	1
1	1	1	0	0

$\Rightarrow$ XOR

(2)

$$a. \quad 5 = 0101, F = 1111, D = 1101, 9 = 1001, 7 = 0111$$

$$0 \times 5FD97 = \underline{010111111011001011}$$

$$b. \quad 6 = 0110, 1 = 0001, 0 = 0000, A = 1010$$

$$0 \times 610A = \underline{0110000100001010}$$

$$c. \quad A = 1010, B = 1011, C = 1100, D = 1101$$

$$0 \times ABCD = \underline{101010111001101}$$

$$d. \quad 0 = 0000, 1 = 0001$$

$$0 \times 0100 = \underline{0000000100000000}$$

(3)

$$a. \quad 0110 = 6, 1010 = A, 1111 = F, 0010 = 2$$

$$\hookrightarrow \underline{0 \times 6AF2}$$

$$b. \quad 1110 = E, 1000 = 8, 0101 = 5, 0001 = 1, 0111 = 7$$

$$\hookrightarrow \underline{0 \times E85517}$$

$$c. \quad 0100 = 4, 1000 = 8$$

$$\hookrightarrow \underline{0 \times 48}$$

(4)

I think is not a good idea.

Because of GIF is limited color palette, and JPEG will loss of picture detail.

(5)

≈ 262144 (bytes)

Ans: $256 \times 256 \times (3+1) \rightarrow$ Because 8 bits = 1 byte

(6) 67 111 109 112 117 116 101 114 32 83 99
105 101 110 99 101

$\hookrightarrow$ Computer Science

(7)

a. $2^2 + 2^0 = 5$ # b. $2^3 + 2^0 = 9$ # c. $2^3 + 2^1 + 2^0 = 11$ #

d. $2^2 + 2^1 = 6$ # e. $2^4 = 16$ # f. $2^4 + 2^2 = 18$ #

(8)

a. $6 = 110$ # b. $13 = 1101$ # c. $11 = 1011$ #

d. $18 = 10010$ # e. $27 = 11011$ # f. 100 #

(9)

a. $4 = 100$, $\frac{1}{2} = .1 \Rightarrow \underline{100.1}$ b. $2 = 10$, $\frac{3}{4} = .11 \Rightarrow \underline{10.11}$
c. $1 = 1$, $\frac{1}{8} = .001 \Rightarrow \underline{1.001}$ d. $\frac{5}{16} = .00101 \Rightarrow \underline{0.0101}$ e. $5 = 101$, $\frac{5}{8} = .101 \Rightarrow \underline{101.101}$

(10)

I: 1, n: 3, A: 4, r: 2, o: 4, d: 1, u: 2
c: 3, a: 2, l: 1, m: 1, p: 1, e: 3, s: 1
' ' = 3, Initial bit = 256 (bits)

next page

Ans: 136 bits